Manufacturing the Third Dimension: Nanofabrication Process Evolution from Planar MOSFETs to Stacked Nanosheet Gate-All-Around Transistors

Abstract:

Historically, Moore's Law followed a two-year transistor density doubling cadence; however, the constraints of economics, thermodynamics and physics have extended this cadence to nearly three years or longer. Therefore, as we continue to scale transistors, we will need advancements in nanofabrication techniques to further reduce dimensions of devices. This report reviews the transition of device architectures based on fabrication improvements from Planar MOSFETs to FinFETs and Gate-All-Around FETs (GAAFETs). We focus our review on the process innovations responsible for transitioning from one architecture to another. Planar CMOS provided the basic framework for fabricating the first CMOS chips using Shallow Trench Isolation (STI); STI replaced LOCOS to prevent bird's beak encroachment into shallow trench isolation and provide sub 250nm isolation. Thermal gate oxidation using the Deal-Grove model defined the Si/SiO2 interface quality at the root of transistor reliability. Anisotropic reactive ion etch (RIE) was used to pattern self-aligned polysilicon gates eliminating gate-to-source/drain misalignment as a potential yield loss mechanism. Once gate lengths fell below forty nanometers due to the limitations of lithographic resolution and short channel effects in MOSFETs, a transition to a three-dimensional structure became necessary. The ability to create FinFETs was made possible due to advances in self-aligned double and quadruple patterning (SADP/SAQP), which extended 193 nm immersion lithography below its resolution limits; high-k/metal gate (HKMG) stacks formed using atomic layer deposition (ALD), replacing the traditional SiO2 with HfO2 and reducing the gate leak current; anisotropic reactive ion etching (RIE) forming vertically oriented fins with high aspect ratios; and highly accurate control of spacers to locate source/drains while minimizing parasitic capacitance in non-planar structures. To achieve the sub-five nanometer node, the GAAFET nanosheet structure was created. Along with the creation of this new architecture, four new fabrication modules were required to have no historical precedent in previously established CMOS architectures. These included: reduced pressure CVD epitaxial Si/SiGe superlattice stack formation defining multiple nanosheet channels; selective channel release etching that achieved SiGe-to-Si selectivity of >100:1 suspending the silicon nanosheets; the formation of inner spacers using gas-phase lateral cavity etching and LPCVD silicon nitride filling to electrically isolate the gate from source/drain areas between sheets; and conformal ALD gate stack formation to form HfO2/TiN layers uniformly across the sub-ten nanometer inter-sheet gaps. In addition to these modules, BDI was also developed to mitigate parasitic sub-fin conduction associated with nanosheet architecture. As each architecturally related advancement has been realized, there has been an increase in fabrication complexity; the number of masks used; tightness of process controls; and introduction of new yield loss mechanisms including: fin height non-uniformity; metal gate granularity induced threshold voltage variability; nanosheet release induced mechanical deformation; and time dependent dielectric breakdown of inner spacers. The progression from two-dimensional planar transistor to three-dimensional transistor represents a fabrication driven approach to overcome fundamental scaling limitations.

Introduction:

The electronics business has been driven by Gordon Moore's prediction from 1965 that the amount of semiconductors per chip will grow by approximately fifty percent roughly every couple of years. This prediction helped sustain growing amounts of transistors per chip over the last several decades as a result of simply shrinking the dimensions of the basic planar Metal Oxide Semiconductor Field Effect Transistor (MOSFET), while increasing the speeds and efficiency of those devices. However, when device gate length fell below forty nanometers, the semiconductor business encountered its first set of physical limits -- namely tunneling and thermal leakage -- along with economic limits -- including exponentially higher cost of lithography equipment, processes, and masks -- as well as thermal limits -- including the need for actively removing heat during the design phase.

These limitations were observed in two forms for the planar MOSFET. First, there is a lithographic limit to how small features can be made and placed correctly. Once the size of gates falls into the range of wavelengths of the light being used to create them, then short channel effects (SCEs) -- including drain-to-source voltage reduction (DIBL), roll-off of threshold voltage, and degradation of sub-threshold slope -- cannot be prevented. Second, as complexity grows in terms of processing requirements for improved performance, reduced power consumption, and high yields, tighter control over all parameters involved in the processing sequence become necessary. These two sets of constraints limited the ability to further scale down the planar MOSFET and created a basis for creating three-dimensional geometries for transistors.

The planar MOSFET fabrication sequence established the key steps used in the fabrication of all subsequent transistor types. At the 0.25µm node, shallow trench isolation (STI) was adopted instead of local oxidation of silicon (LOCOS) so that the lateral "bird's beak" encroachment characteristic of LOCOS was removed. In addition to providing a means to isolate adjacent device elements electrically, STI also provided the first opportunity to apply CMP to achieve surface planarity. Thermal gate oxidation using the Deal-Grove linear-parabolic rate law formed the Si/SiO2 interface that provides the electrical properties for the MOSFET. The self-aligned polysilicon gate structure formed using anisotropic reactive ion etching (RIE) allowed elimination of misalignment errors that occur when forming gates over sources and drains. It also permitted extremely small channel lengths to be obtained. Taken together, these techniques provided a fabrication sequence that supported planar CMOS from micron-scale devices to those scaled down to less than 100nm.

Transition to three-dimensional architectures did not represent a single design decision but rather a necessity dictated by fabrication capability. Once it became impossible to provide sufficient electrostatic control of the channel region through a single-sided gate as channel lengths decreased and leakage current began to dominate device behavior, a second-generation architecture was needed. The FinFET represented one solution to this problem. By placing the gate on three sides of a vertically oriented silicon fin, the FinFET restored electrostatic control of the channel region. This required an entirely new set of fabrication capabilities. Self-Aligned Double Patterning (SADP) and Self-Aligned Quadruple Patterning (SAQP) were developed to enable extension of 193nm immersion lithography past its fundamental resolution limit so that fin pitch dimensions accessible only with multi-patterning methods could be defined. High-K/Metal Gate (HKMG) stacks formed by ALD were used in place of thermally grown SiO2 to minimize gate-leakage as equivalent oxide thicknesses approached values <1nm. Anisotropic RIE was used to form high-aspect-ratio vertical fins with near-vertical sidewall profiles and

precise spacer engineering was used to define source/drain placement essential to controlling SCEs while minimizing parasitic capacitance in non-planar geometries.

Although electrostatic advantages existed for the FinFET, fabricating smaller widths for further scaling at sub-7nm nodes resulted in additional fabrication difficulties that eventually constrained how far the FinFET technology could be scaled. A third-generation architecture was therefore introduced. In this Gate-All-Around FET (GAAFET), the gate completely surrounds the channel on all four sides through a stack of horizontally suspended silicon nanosheets. Fabrication of GAAFETs necessitated process modules never before seen in prior CMOS manufacturing. Formation of epitaxial Si/SiGe superlattices by reduced-pressure CVD defined the alternating sacrificial and channel layer materials from which the nanosheet stack is fabricated. Selective channel release etching with SiGe-to-Si selectivity >100:1 suspend silicon nanosheets in free space for gate encapsulation. Inner spacer formation via gas-phase lateral cavity etching and LPCVD silicon nitride fill isolates the gate from source/drain regions between each nanosheet. Uniform conformal ALD gate stack deposition achieves gap-filling coverage within sub-10nm inter-sheet gaps that no line-of-sight deposition method could reach. Finally, bottom dielectric isolation (BDI) was added to reduce parasitic conduction pathways that exist uniquely in nanosheet-based structures.

This report will discuss each of these three architectural generations in historical order, emphasizing only fabrication process innovation responsible for each transition. Reference will be made to device physics and circuit level performance only as necessary to describe the specific fabrication requirements for each architecture. Each discussion will cover only the primary nanofabrication techniques employed in each architecture; only those integration issues introduced at each node relevant to describing fabrication challenges; and only those aspects of yield impact associated with each module described above that relate to fabrication.

Planar MOSFET/CMOS

Overview:

Understanding the progression from planar CMOS technology to FinFET and GAAFET transistors requires an appreciation for how each of the fabrication technologies listed (STI, gate oxidation, polygate patterning) provided a solution to critical issues with respect to scaling or reliability at their respective times of development and have since become fundamental building blocks for all advanced transistor process architectures.

Shallow Trench Isolation (STI)

Why was it needed:

Before STI, device separation in planar CMOS had been accomplished using Local Oxidation of Silicon (LOCOS). “In LOCOS, silicon nitride is patterned above the region where an active area exists. A relatively large field oxide is then thermally developed in the un-protected areas. The major limiting factor associated with LOCOS is known as the “bird’s beak.” This phenomenon occurs when the oxidant will diffuse under the nitride edges after the nitride has been formed as the field oxide is being produced. Therefore, the thickness of the field oxide can grow or become thicker in the direction perpendicular to the substrate and also increase its lateral thickness toward the center of the transistor structure.”[1] Although it may seem minor at first glance, this lateral growth in thickness caused many problems as devices were designed to

pack more tightly together than previously. The packing density requirement for future generations of devices required that each generation of devices have a lateral dimension size reduced by approximately 30% compared to the previous generation. Consequently, lateral expansion of the field oxide created by LOCOS became so severe that there was no longer room left on either side of the transistor to isolate them from adjacent transistors. Thus, the need for very small lateral dimensions resulted in a serious conflict between the desire to achieve higher packing densities and the amount of space needed to isolate adjacent transistors. With this conflict came a second problem with LOCOS. It was difficult to develop a flat surface using LOCOS. The field oxide grown using LOCOS results in a non-flat surface, creating difficulties in achieving focus at different depths for all of the succeeding photolithographic processes. "When the dimensions of devices began shrinking below microns and as lithographic requirements became tighter, surface non-planarity of the field oxide grown using LOCOS became completely incompatible with the submicron lithographic requirements." [2]

Process flow and nanofabrication techniques:

Stage one is initiated when a thin layer of SiO2 (5-15 nm) is formed on the silicon substrate due to thermal oxidation. Subsequently, a layer of silicon nitride (Si3N4) is deposited via low pressure chemical vapor deposition (LPCVD) to a thickness of about 150 nm. The nitride will serve as both the CMP stop layer and will protect the silicon from being oxidized during the subsequent thermal liner oxide oxidation. The nitride can be made to be of varying thickness depending upon several variables including: the final trench depth, CMP selectivity, and topography (step height) of the isolation. After the nitride has been deposited onto the silicon substrate it is lithographically defined and subjected to anisotropic reactive ion etching (RIE) in order to create the trench patterns throughout the nitride/oxide stack down to the bottom of the trench in the silicon substrate. Typically, the depth of these trenches' ranges from 250-400 nm. Once the oxide pad has been etched, a thermal oxide (the liner oxide) is thermally oxidized in the trench. One difference between STI and LOCOS is that the thermal oxidation of the liner oxide stops once the thin liner oxide is formed and a deposited oxide fills the remaining portion of the trench. The primary purpose of growing the liner oxide within the trench is to round off the sharp corner edges of the trench. If the corner edges of the trench are not rounded, they may become electric field concentration points that could degrade the reliability of the gate oxide at the trench edge. The trench fill oxide is deposited using TEOS-based chemical vapor deposition (CVD), either LPCVD TEOS or high-density plasma CVD (HDP-CVD). "HDP-CVD processes have shown successful use in filling gaps as wide as 100 microns using aspect ratios greater than ~3:1." [3] Once the fill has been completed, CMP is done until it stops on the silicon nitride layer thereby flattening out the oxide layer back to where it meets up with the nitride. Once the CMP is complete, the silicon nitride layer is removed, usually in a mixture of phosphoric acid, hydrogen peroxide and water at 80 degrees Celsius. Prior to removing the nitride layer there is sometimes a "deglaze" step that takes place prior to nitride removal. A dilute HF etch removes any surface oxides prior to stripping away the silicon nitride layer. After removal of the nitride layer, a dilute HF solution is also applied to strip away any remaining buffer oxide. A very thin layer of screen oxide of 5-20nm thickness is then grown on the exposed silicon surface. [4][5]

Technological advancement and yield impact:

STI has an important benefit (as compared to LOCOS) due to its low lateral encroachments, which enable it to potentially scale down to deep submicrometer generation

technologies. In addition, STI has the ability to produce a surface which can be significantly more flat as a result of lateral isolation than surfaces produced using techniques based on LOCOS. One key enabling factor that enabled STI for large volume manufacturing was CMP. It would have been impossible to achieve the degree of uniformity necessary to allow overlay tolerances of sub-0.25 micrometer lithography without CMP. The two primary issues related to yield of CMP for STI are: polishing rates depending on the pattern density; and corner erosion of nitride. There were two critical requirements for planarizing STI: the removal of all CVD trench fill oxide that had been deposited in the active areas, and there could be no erosion of the nitride over the active silicon regions. As a result, a variation in pattern density from 40% to 60% alone would cause a variation in nitride thickness of approximately 32 nanometers. This problem of loading patterns during CMP (wherein densely populated regions polished at a greater or lesser rate than sparsely populated regions) added another layer of wafer-level uniformity to be solved through slurry chemistries being optimized, pad conditioning control, and dummy fills being inserted into layout design rules. [5]

Gate oxidation

Why it was needed:

Silicon dioxide is the most important and critical dielectric material used in making planar MOSFETs. Silicon dioxide physically separates the gate metal from the silicon channel but allows the silicon dioxide to modulate the silicon channel's carrier concentration using the electric field. The silicon dioxide is produced by oxidizing the silicon in the channel at a relatively low temperature (1000°C) until it produces a very thin insulating film. The Deal Grove Model describes how the silicon dioxide forms through self-limited oxidation. Because of the importance of both the quality of the silicon dioxide itself and its interfacial relationship with the silicon in controlling carrier mobility, leakage current, threshold voltage and ultimately long-term reliability of devices, the presence of an impermeable, strong thermal oxide is arguably responsible for why silicon has been used in semiconductor processing and manufacturing for so many years. Germanium and Gallium Arsenide are both better electrical materials than silicon, but no one has developed a good insulating oxide on either germanium or gallium arsenide that could be grown by simply thermally oxidizing the substrate in a large batch process.

Process flow and nanofabrication techniques:

Dry oxidation is typically used to grow very thin oxides due to its ability to reduce interface traps. In contrast, wet oxidation is commonly used for thicker oxides as it has a faster rate of growth; however, this process can create a larger number of dangling bonds at the oxide-silicon interface, resulting in more interface trap states. Additionally, wet oxidation produces a less dense oxide than dry oxidation, thereby reducing the dielectric strength. A common method of producing thicker oxides is to use a wet oxidation step in the middle of a three-step dry-wet-dry oxidation cycle. The initial and final steps in the cycle use dry oxidation and therefore provide a good quality oxide surface on both sides of the thick oxide. The Deal-Grove model describes how thermally grown SiO_2 grows according to the Deal-Grove linear-parabolic rate law. This model was originally developed in 1965 and proposed that the oxidation rate of SiO_2 is controlled by the rate of oxygen diffusing through the already-grown oxide film (the parabolic term), and by the chemical reaction of oxygen reacting with silicon at the oxide-silicon interface (the linear term). These two terms combine to form the classical linear-parabolic rate law. Thermal gate oxidation is generally completed prior to the doping of the substrate where the

source and drain are located. It is impossible to perform thermal oxidation after the source and drain have been doped since doing so will destroy the placement of the dopant atoms. [6][7][8]

Technological advancements and yield impact:

Gate oxidation wasn't just a way to create gates it defined how the MOSFET works as a device. However, as we have scaled the MOSFET below 130 nm, the gate oxide has become the limiting factor. Continued scaling has reduced the thickness of the gate oxide to where direct tunneling is now a significant problem. Alternate high-k dielectrics are currently under consideration as replacements for SiO2 with the hope of achieving increased capacitance while controlling gate leakage. Today, the state-of-the-art oxides are approximately 1.5 nm thick, or roughly five atomic layers: including two SiOx layers for attachment to each side of the silicon substrate, and the polysilicon gate electrode. It will likely be possible to continue thinning down the SiO2 layer but once direct tunneling becomes a problem, it will no longer act as an insulator. In terms of yield, gate oxide integrity (GOI), is probably one of the largest yield determinants in the manufacture of CMOS devices. A defect density in the gate oxide will determine what percent of transistors on a chip will fail TDDB tests. Any particulates or contaminants present during processing can cause defects within the gate oxide layer. Surface roughness caused by previous process steps can also affect GOI. Metallic impurities also decrease GOI yield. Both Rapid Thermal Processing (RTP) for gate oxidation and In-Situ Steam Generation (ISSG) oxidation at sub-100 nm nodes have improved oxide uniformity across the entire wafer when compared to traditional batch furnace processes. ISSG has improved the rounding of corners at trenches and the oxide quality adjacent to shallow trench isolation (STI) which were major contributors to GOI yield loss. [9]

Polysilicon Gate Patterning

Why it was needed:

Early MOSFET manufacturing used aluminum gates. Aluminum has a melting point of 660°c. Therefore, an aluminum gate can't be made until all of the high temperature processes have been completed. Since the gate must be made before the source/drains, this results in two different layers (gate/source/drain) with the same or nearly the same x-y coordinates. If these two layers are misaligned, there will be variations in parasitic overlap capacitance and sometimes it may not even function. The development of the self-aligned gate is critical to advancing MOSFET technology. “Researchers found that heavily doped polysilicon was sufficiently electrically conductive to eliminate the need for aluminum. Therefore, the gate layer can be developed anywhere throughout the many step process. To develop a self-aligned gate, the first step is to create the gate dielectric material early in the process. Then a layer of the gate material is created and developed into shape on top of the gate dielectric material. Finally, the source and drain areas are doped. The area where the source and drains are patterned represent only the outer boundaries of the source and drains; therefore, the inner boundary is shielded from the dopant by the gate itself. Because of this, the source and drains will self align themselves to the gate and when they do, they will always be properly aligned. There is no need to build a wider gate than desired, thereby dramatically decreasing parasitic capacitance.” [10] The use of self-aligned gates in MOSFETs is one of several innovations that contributed to the exponential growth of computing performance in the 1970's. By utilizing the gate as a mask for developing the source and drain diffusions, you simplify your process and improve yield.

Process flow and nanofabrication techniques:

"After you oxidize the gate, an undoped polysilicon film with a thickness ranging from 150-300 nm is deposited via low pressure chemical vapor deposition (LPCVD) from silane (SiH_4) at temperatures about 620–640° C. Following this, the film is doped – usually through phosphorus oxychloride ($POCl_3$) diffusion to create n-type gates" [11]; although there are other applications where the film is doped in situ. This creates the desired sheet resistances in the polysilicon that will be used as the gate electrodes. Silicon dioxide or silicon oxynitride layers are commonly applied over the polysilicon to function as etch masks to define the location of the gate cap. In addition, these layers will function as a CMP stop layer when ILD planarization occurs to prevent excessive removal of the silicon dioxide underlayers. Photolithography defines the gate length which is the most important feature in all steps of the flat CMOS processing sequence since the electrical channel length of each transistor on the chip is directly dependent upon the gate length. Polysilicon is then etched using chlorine based RIE plasma chemistries ($Cl_2/HBr/O_2$). The RIE plasma chemistries provide excellent selectivity to the underlying gate oxide so that the etch can terminate at the gate oxide surface and therefore prevent etch into the channel silicon located beneath. Standardly, the gate electrode is heavily doped polysilicon that has been defined by anisotropically etched reactive ion etching. Junction extensions have been created by ion implanting with the polysilicon gate functioning as an implant mask. Since no additional masking step was required to define the locations of the junctions, this method is referred to as self-aligned polysilicon gate technology.

Technological advancement and yield impact:

An advantage of using self-aligned polysilicon gates was elimination of another major category of defects associated with misalignment during fabrication of aluminum-gate CMOS. With the gate serving as the implantation mask for the source/drain areas, there will be no additional failures due to misregistration or overlap between the gate and source/drain implants and transistors will have significantly less variability in their effective channel length. Therefore, because of the significant reduction in threshold voltage variability among all transistors, parametric yields improve and smaller minimum features are possible. Yield loss from polygate processing occurs primarily through two mechanisms: critical dimension (CD) control of the gate, and gate line-edge-roughness (LER). Since each transistor's effective channel length is determined by how well-defined the dimensions of the gate are, a 1-nm deviation in the gate's critical dimension results in a corresponding difference in threshold voltage and drive current. "To achieve acceptable uniformity in gate-critical dimension at sub-100nm gate-lengths, OPC (optical proximity correction) was incorporated into photomasks and PSM (phase-shift masks) were developed to correct for CD variations due to diffraction. For example, at the 45 nm technology generation, polysilicon gates were replaced with metal gates, such as titanium nitride, to eliminate the depletion of polysilicon that reduces gate capacitance and increases EOT. A new replacement-metal gate process was developed, which consists of forming a temporary dummy polysilicon gate to define the source/drain junctions and then selectively removing this gate and replacing it with a metal gate stack." [12]

FinFETs

Fin Patterning: SADP and SAQP

Why was it needed:

“EUV lithography wasn’t ready for production when the semiconductor industry reached the 20 nm process node. Traditional single exposure 193nm immersion lithography cannot accurately create the layouts required to get to those sizes.” [13] Therefore, the industry developed self-aligned multi-patterning techniques, including SADP (self-aligned double patterning) and SAQP (self-aligned quadruple patterning), to achieve the necessary layout resolution.

Self-Aligned Double Patterning (SADP):

Self-Aligning Double Patterning is a method of fabricating transistors that uses one lithography exposure to double the density of features. It involves forming spacers on a mandrel, where the spacers are the final mask and will provide better alignment than traditional multi-exposure methods. The SADP process flow is as follows:

1. Mandrel Formation,
2. Slimming (Optional),
3. Deposition of Spacers,
4. Etch back,
5. Removal of the Mandrel,
6. Final Etch.

In my presentation I showed an example of how SADP works; the orange boxes represent the mandrel that will determine the size and shape of the features. The green box represents the conformal deposition that may include atomic layer deposition (ALD), chemical vapour deposition (PECVD, LPCVD, CVD), or plasma-enhanced ALD. The deposition is then etched back so that the mandrels and sidewalls remain. The mandrel is then removed from the substrate, and the sidewalls remain. A final etch is performed so that the "Fins" remain on the substrate. The mandrels for advanced nodes are typically amorphous carbon, amorphous silicon, poly silicon, or silicon oxide. For conformal deposition, ALD is the preferred method due to the fact that it provides good conformity, operates at low temperatures, and has excellent thickness control. For the etch back, RIE is the preferred method due to the fact that it produces highly anisotropic etches that produce vertical profiles, has good selectivity for the spacer material, reduces edge placement errors, and minimizes line edge roughness. Depending upon the type of mandrel material, ashing, wet etching, or dry etching is used for mandrel removal. [14][15]

The main benefit of SADP over LELE (litho etch litho etch) is the inherent alignment of the two patterns. Because the two patterns are produced in the same process, they inherently align. The largest challenge with SADP is the etch bias on the spacers. If the spacers are etched too much, they become thinner, which creates a width bias between the adjacent shapes on the substrate.

“Device fabrication begins with the use of 193nm water immersion lithography to pattern the substrate. Due to the 45nm half-pitch resolution limit of 193i lithography” [16], SADP is used to create fins less than 40nm wide. “For inter-fin isolation, shallow trench isolation (STI) is created by depositing SiO2 using low-pressure chemical vapour deposition (LPCVD) or flowable chemical vapour deposition (FCVD) with TEOS as the precursor.” [17] These methods provide good conformality and void-free gap filling at high aspect ratios. “After deposition, the STI oxide is planarized using chemical mechanical polishing (CMP). The STI oxide is then recessed using either dilute hydrofluoric acid (dHF) or buffered hydrofluoric acid (BHF) wet

etching to expose the fin to the desired height." [18] The recess step is critical due to the potential for different etch rates between densely populated regions and isolated regions. Variations in fin height across the substrate can lead to differences in threshold voltage, and ultimately to lower parametric yield. For advanced nodes less than 10nm, dry plasma-based recess processes are implemented to further improve fin height uniformity compared to wet etching alone.

Self-Aligned Quadruple Patterning (SAQP):

By repeating the SADP process, an additional halving of the pitch is possible. This is known as Self-Alignment Quadruple Patterning (SAQP). With SAQP, the pitch is decreased to 19nm, enabling the creation of fins that are 5-7nm wide. "SAQP is similar to SADP in that the spacer deposition and etch cycles are repeated a second time around the spacers from the first cycle. The second cycle of spacer deposition and etching quadruples the feature density from a single lithographic exposure." [19][20] This significantly increases the complexity of the process, the number of masks, and the cycle time, but it also makes possible the creation of fin pitches that are not achievable by any other UV-based method.

Impact on Yield of SADP and SAQP:

"The primary yield issues with SADP and SAQP are spacer non-uniformity (line edge roughness and critical dimension uniformity). Additionally, mandrel removal selectivity, cut mask overlay errors, and variations in ALD spacer thickness all contribute to fin width variability, which results in a shift in the threshold voltage across the die. Since SAQP adds an additional patterning cycle to the process, these errors accumulate. Therefore, it is imperative that there is tight process control at each of the deposition and etch steps." [21][22] RIE has higher anisotropy characteristics compared to competing technologies, which is beneficial in device fabrication. The anisotropic nature of RIE enables greater accuracy in the resultant device dimensions, resulting in cost savings in the development phase as well as increased production yields.

High-K/Metal Gate

As the gate oxide thickness began to scale below 2nm in SiO2-based planar MOSFETs, quantum tunnelling began to cause gate leakage currents. This resulted in accelerated device degradation, thermal stress, and shortened lifespan. By replacing SiO2 with a high-k dielectric material, it is possible to create a physically thicker gate insulator while maintaining the same equivalent oxide thickness. This effectively eliminates current leakage while preserving capacitance. Concurrently, the polysilicon gate was replaced by a metal gate. This eliminated poly depletion effects, including increased gate depletion, threshold voltage shifts, performance degradation, gain/linearity loss, and scaling limitations. The first high k/metal gate devices were released to the market in 2007. This represented the return of metal gate technology on silicon after a hiatus that occurred in the early 1970's when polysilicon gates were utilized in all devices. Intel was the first to utilize high k/metal gates in its 45nm products.

Materials:

"The most commonly used high-k dielectric is Hafnium Oxide (HfO2), which has a dielectric constant of approximately 20-25, versus 3.9 for SiO2." [23] "Alternative high-k dielectrics include Zirconium Oxide (ZrO2,k~23), Aluminum Oxide (Al2O3,k~8-9), and

Titanium Oxide (TiO2,k~20-80).” [24] An interfacial layer of SiO2 or SiON is typically grown between the silicon layer and the high-k dielectric. This is done to preserve the integrity of the interface and to minimize the density of defects at the interface. Defects at the interface can cause reduced mobility, unstable threshold voltages, leakage current, and degraded device performance. “The metal gate materials that are typically chosen for their compatibility with high-k dielectric and their ability to tune the effective work function for both nMOS and pMOS devices include Titanium Nitride (TiN), Tantalum Nitride (TaN), and Aluminum (Al). Other metals that can be used for metal gates include Titanium/Titanium Carbide (Ti/TiC), Tantalum (Ta), Ruthenium (Ru), and Tungsten (W).” [25] Each of these metals has unique work function properties and threshold voltage tuning capabilities that can be utilized to define complex FET specifications.

Gate First vs Gate Last Integration Approaches:

There are two common process flows that are used to pattern a high k/metal gate stack: Gate First and Gate Last. “The Gate First approach is very sensitive, since the capping metal that must be patterned is very thin. Photoresists do not adhere well to the metal layers, and this is exacerbated by the wet chemicals that are used to etch the capping metal. In the Gate Last approach, a sacrificial polysilicon layer dummy gate is first patterned and serves as a placeholder throughout the high-temperature source/drain annealing steps. Once the source/drain activation anneal is complete, the dummy gate is removed by selective etching, and a high-k dielectric and metal gate stack replaces it. The Gate Last approach protects the thermally sensitive metal gate materials from the high-temperature processing done by annealing.” [26]

Techniques Required for NanoFabrication:

Atomic Layer Deposition (ALD) is a critical technique used for high k/metal gate FinFET architectures. In conventional high k/metal gate processes in MOSFETs and early FinFETs, sputtering was used, but it did not provide face conformality when patterning 3D channel geometries. Therefore, ALD was used to achieve face conformality, angstrom-level thickness control, and uniform coverage over the high aspect ratio three-sided fin surface. Techniques used in conjunction with ALD include CMP for planarization, selective RIE for dummy gate removal, and wet chemical cleaning.

Yield impact:

For more advanced nodes, the polycrystalline nature of the metal gate leads to threshold voltage variations across devices, which are large enough to be considered yield limiting factors. Variations in interface trap densities also contribute to variability in threshold voltage. Other processes have similar precision requirements such as CMP to leave a “dummy” gate exposed without damaging nearby structures.

3D Etching techniques

Why was it needed:

High-aspect-ratio vertical silicon Fins with heights of 50 – 100 nm and widths less than 10 nm are formed through FinFET fabrication. These dimensions require etching of near-perfect accuracy and minimal surface roughness. Wet etches (both isotropic and moderately anisotropic) are inadequate for this application because they would laterally undercut the fin geometry.

Reactive ion etching (RIE) for Fin Formation:

Plasma and ion bombardment remove material in RIE, creating a high-aspect-ratio anisotropic etch profile. "A plasma is created from reactive gas chemistries in RIE such as fluorine (F), chlorine (Cl), and bromine (Br). Examples include sulfur hexafluoride (SF6), octafluorocyclobutane (C4F8), chlorine (Cl2), hydrogen bromide (HBr). Additive gases are sometimes added to enhance profile definition and provide passivation. Oxygen (O2) and argon (Ar) are some common additives." [27] Reactive gases are introduced into a vacuum chamber and excited by RF fields to form a plasma. The reactive gases are broken down into ions and radicals within the plasma. The ions and radicals are then directed toward the surface of the substrate, tearing molecules away from the substrate to produce an anisotropic vertical etch while passivating the sidewalls using products of the reaction, thereby protecting them from lateral attack.

The hard mask stack used above the fin (typically SiO2/SiN bilayer) transfers the SADP/SAQP defined pattern into the silicon substrate. Nitride spacers can be used as hard masks for the reactive ion etch of silicon due to their high etch selectivity, anisotropic etching characteristics and effectiveness at resisting attacks from forms of etching. "Vertical silicon fins 690nm high, processed with an aspect ratio of 29:1 and smooth sidewalls, were achieved on <110> bulk silicon wafers when RIE and wet etching with tetramethylammonium hydroxide (TMAH) were applied for planarized oxide trench isolation of the base of the etched fins. This opens the possibility of processing FinFETs with different channel widths." [28]

Technological advancements and yield impact:

"The development of high-density plasma RIE tools with inductively coupled plasma (ICP) sources allowed the independent control of plasma density and ion energy, enabling high etch rates, high anisotropy, and low surface damage." [29] Decoupling was necessary to achieve the fin profile uniformity required for adequate threshold voltage stability among tens-of-millions of fins per die. Fin line-edge roughness (LER), caused by both lithography and etch processes, is primarily responsible for yield loss. Variation in LER directly results in variation in fin width and since fin width determines the transistor body, any variation will result in variation in threshold voltage and therefore a reduction in parametric yield. By introducing RIE, we solved one of the major contributors to fin variation and thus open additional avenues for reducing yield loss.

Spacer Engineering

Why was it needed:

In FinFET, the spacer is a dielectric layer deposited on the sidewalls of the gate structure between the gate, the source and drain epitaxy regions. It serves several critical functions: it defines the offset between the gate edge and the source/drain junction (controlling parasitic series resistance and overlap capacitance); it shields the gate stack from subsequent etching and ion implantation processes; and it acts as a hard mask during SADP/SAQP patterning.

Process flow:

"After gate patterning has been done, a conformal dielectric film, which is typically silicon nitride (Si3N4) or a low-k nitride variant is deposited over the gate stack and fin using

CVD or ALD. An anisotropic RIE etch (spacer etch) then removes the horizontal portions of the film, leaving spacer material only on the vertical sidewalls of the gate." [30] An accurate anisotropic and selective spacer etch clears the top and sides of the fins, leaving spacer only on the sidewall of the gate, inducing minimal damage on the fin and top hardmask (if present). However, the actual spacer etch introduces challenges such as thinning of the spacer, recess of the fin, and shoulder pulldown. In order to obtain a reliable device, these effects must be minimized by a spacer etch process with enhanced anisotropy and selectivity to the fin. The spacer width directly controls the source/drain extension junction placement, which affects both the short channel control and the access resistance. In advanced nodes, multiple spacer layers (inner and outer spacers of different dielectrics) are used to tune these tradeoffs independently.

NanoFabrication techniques required:

To perform spacer engineering requires both conformal CVD or ALD for even deposition throughout 3-Dimensional fin and gate architecture, as well as highly selective anisotropic RIE. To maintain selectivity between spacer material, underlying fin/gate-cap/STI-OXIDE must be held to a fine degree. "Low-k dielectric spacers (SiOCN) are utilized for advanced FinFET nodes to minimize parasitic gate-to-source/drain capacitance which directly impacts device speed." [31]

Technological advancements and yield impact:

Spacer engineering enabled the self-aligned source/drain formation process that is essential to all modern sub 20nm transistors. "Without precise spacer control, the source/drain epitaxy would encroach under the gate or be placed too far from the central channel, degrading both speed and leakage." [32] Thickness variation of spacers is another major yield factor: If spacers become too thin, they may lead to gate-to-drain shorts whereas if spacers become too wide they increase access resistance decreasing drive current. "The move to use low-k spacer materials at 10nm and below required development of new plasma chemistries. This was due to two main issues: prevention of carbon deposition (traditional O2-based plasmas strip out vital carbon-containing molecules which subsequently allow water absorption onto the dielectric leading to property changes.) As mentioned above other reasons for needing these new chemistries include anisotropy control, achieving higher selectivity, minimizing material damage." [33]

Why GAAFET Was Necessary

Scaling FinFETs beyond the 7nm node results in severe short-channel effects, which motivated a move from a tri-gate architecture to a gate-all-around architecture. Among the gate-all-around architectures explored by the semiconductor industry, wider nanosheets provide higher on-current and improved electrostatic control over FinFETs.

Epitaxial Si/SiGe Superlattice Stack Formation

Why was it needed:

In a FinFET, the channel is defined by a single vertical fin of silicon. In a Gate All Around transistor, the channel consists of multiple horizontally stacked silicon nano sheets, each fully surrounded by the gate. The only practical method for forming these precise multilayer stacks with atomic-scale thickness is epitaxial growth. This represents a fundamental departure

from all previous CMOS fabrication, which relied on patterning a single bulk or SOI silicon substrate rather than growing an engineered multilayer heterostructure from the ground up.

Process flow and nanofabrication techniques:

"The SiGe/Si superlattice is grown epitaxially on silicon wafers via low pressure chemical vapor deposition (LPCVD) for GAAFETs. At temperatures <650°C, the SiGe/Si superlattices are produced with precise layer thicknesses and with well-defined germanium content. Precursor chemicals used in LPCVD include dichlorosilane (SiH2Cl2), commonly abbreviated DCS and germane (GeH4) for the precursors to silicon (Si) and germanium (Ge), respectively." [34] Approximately 8-10 nm thick alternating layers of crystalline silicon and silicon germanium are epitaxially grown. At this point, two parameters are measured: layer-by-layer thickness and the amount of germanium present in each layer. Material uniformity, layer uniformity, and superlattice quality are essential in establishing and maintaining the channel dimensions and electrical properties and failure to achieve one or more of these could result in devices that perform poorly or significantly reduce yields. Each material in the stack serves a particular purpose. Silicon layers provide the active channel nanosheets that carry current. Silicon germanium layers act as sacrificial spacers providing the lateral separation between the channel regions that can be selectively removed later in the channel release step. "The amount of germanium in the silicon germanium layers (typically 20-30 atom percent) is determined to optimize etch selectivity when removing the silicon germanium layers relative to the remaining silicon layers." [35] Lattice misfit strains generated by the different materials may generate dislocations. However, since the in-plane strain varies throughout the stack with compressive stresses developed in the silicon layers and tensile stresses developed in the silicon germanium layers, there is limited stress build-up in either layer. Following the superlattice formation, the stack is processed similarly to FinFETs utilizing SADP and SAQP lithographic techniques described for FinFETs, followed by anisotropic RIE to etch the entire multilayer stack into individual fins. Each fin now contains the complete nanosheet stack rather than a single silicon body.

Technological advancement and yield impact:

The channel release etch is arguably the most yield sensitive processing step in producing GAAFETs. Any under etching of the silicon germanium layers would leave residual silicon germanium that would prevent contact between the gate electrode and the nanosheet along its entire length resulting in degradation of electrostatic coupling. Conversely, any over etching would remove some portion of the silicon nanosheet reducing its width and contributing to parametric yield loss. If left unresolved, mechanical deformation problems will cause variability in nanosheet spacing creating variations in dielectric fill quality within the gate gap between adjacent nanosheets and generating threshold voltage differences between adjacent nanosheets. To address the issue of developing process control metrology capable of confirming complete removal of the silicon germanium layers while confirming integrity of the silicon nanosheets were necessary to implement a multi-step etch strategy.

Inner Spacer Formation

Why was it needed:

The inner spacer was created for two reasons: To provide electrical isolation between the gate and the source/drain areas to prevent shorts through the gap formed between adjacent nanosheets. To decrease the parasitic capacitance associated with the gate-to-source capacitance. The dielectric spacer that separates each SiGe sacrificial layer from the future source/drain epitaxy is not found in a FinFET and therefore inner spacer formation is a completely new and significantly more complicated process module. The creation of the inner spacer involves recessing the outer parts of the SiGe layers within the multi-layered structure, utilizing a lateral etch method. Following this, the resulting recesses are then filled with dielectric. Inner spacer integration is the most complicated process module within the nanosheet fabrication process. The process of creating an inner spacer requires a very high degree of selectivity in addition to a high degree of precision in controlling lateral etching.

Process flow and nanofabrication techniques:

"The inner spacer process flow proceeds as follows. the sequence is hard mask deposition; source/drain fin recess to open the active area and expose the SiGe layer ends; lateral isotropic SiGe cavity etching to define the position and size of the inner spacer; conformal inner spacer dielectric deposition; controlled anisotropic etch back of the spacer film to form the final inner spacer geometry." [36] The cavity etch selectivity requirement is extreme. The etching process demands an ultra-high selectivity of SiGe and Si to avoid Si damage to either layer. Otherwise, thinning or roughening of the channel layers would degrade carrier mobility, and the quality of source/drain epitaxy would be impacted by the damaged end of the Si nanosheet, raising contact resistance. "There are three dry etching methods under consideration for use in etching the cavity of each step: Inductively Coupled Plasma (ICP); Remote Plasma Source (RPS); and Gas Phase Etching." [37] Of these three options, gas phase etching exhibits a selectivity for SiGe sacrificial layer that is higher than both ICP and RPS. Thus, gas phase etching appears to be the best option for use in reducing damage to the Si nanosheet while etching inner spacer cavities. "The selectivity ratio for SiGe/Si is approximately 9 times greater when using gas etching versus ICP, and 2 times greater than when using RPS." [38] LPCVD Silicon Nitride is also considered to be one of the best candidates for the dielectric fill material. "LPCVD silicon nitride provides a good film filling effect; using ICP with a combination of CH2F2/CH4/O2/Ar allows for preparation of a precisely controlled inner spacer structure. The selectivity ratio achieved by silicon nitride inner spacer etch is >100:1 vs. Si and >30:1 vs. SiO2." [39]

Reliability and yield impact:

"Time-Dependent Dielectric Breakdown (TDDB) is a major reliability concern specific to nanosheet architectures. The primary cause of this problem is largely attributed to difficulty in controlling the thickness and morphology of the inner spacer combined with necessity to decrease the thickness of the inner spacer and lower dielectric constant of the spacer material in order to improve device performance. The moon shaped profile of the inner spacer present in nanosheet devices presents a much greater risk for TDDB and current leakage relative to traditional top spacers. Increased focus has been placed on improving both process development efforts aimed at developing more square-shaped inner spacers with improved uniformity. Uniformity improvements result in both enhanced reliability and performance." [40]

Conformal ALD for Gate Stack Deposition

Why was it needed:

The silicon nanosheets after channel release are suspended in free space and are thus exposed on all sides via the narrow intersheet gaps. Therefore, the gate dielectric and metal gate need to be applied to all surfaces of all of the nanosheets in the stack (both the bottom surface of every sheet and the top surface of the sheet underneath.) A three-dimensional deposition problem that is physically impossible using any line-of-sight method like PVD sputtering. Only ALD could provide a way to conformally cover the fully enclosed, high aspect ratio, <10 nm intersheet spacing.

Process flow and nanofabrication techniques:

"ALD operates through self-limiting surface reactions. A metal precursor is pulsed into the chamber and reacts with the available surface sites. The chamber is purged, then an oxidant (typically water vapour or ozone) is pulsed to complete the reaction and regenerate surface sites. This cycle deposits precisely one atomic monolayer per cycle, regardless of surface geometry. The deposition rate is typically 0.5-1.5 angstroms per cycle." [41] The SiGe is removed using channel release etch once the inner spacers have been inserted. In the small spaces between the silicon nanosheets, ALD deposits the gate dielectric and metal. conformal HfO2/TiN/W deposition is performed with specific attention paid to the bottom nanosheet to maintain good electrostatic integrity. the gate stack deposited in a GAAFET is structurally the same as the HKMG stack described for FinFETs, a thin SiO2 or SiON interfacial layer, followed by HfO2 high k dielectric, followed by metal work function layers (TiN for PMOS, for NMOS a n-type function metal is chosen), but the conformality requirement is far more stringent due to the enclosed nanosheet geometry. A critical constraint unique to GAAFET is the limited space between nanosheets, which is typically only 8-10nm after accounting for nanosheet thickness. The cumulative thickness of the gate stack (interfacial oxide + HfO2 + work function metal) must fit within this gap while leaving a void for a low resistance fill metal (typically tungsten or cobalt). when ALD thickness is non-uniform or excessive, the intersheet gap is not fully filled, creating void defects that compromise gate contact and threshold voltage uniformity.

Technological advancement and yield impact:

The use of ALD for gate stack in a GAAFET represents the most advanced application of the technique in mainstream CMOS. The challenge is not only conformality but also step coverage uniformity across the full depth of a three-stack structure, where the bottom of the nanosheet may receive slightly fewer precursor molecules per cycle than the top nanosheet due to diffusion limitations in confined spaces. This creates a top-to-bottom variation in gate dielectric thickness, which translates directly to threshold voltage variation between the nanosheets within a single device. This device variability did not exist in previous transistor architecture such as MOSFETs and FinFETs. This variability is an active area of process development and is managed through precise ALD cycle timing, precursor dose control, and elevated deposition temperatures to improve surface diffusion.

Bottom Dielectric Isolation (BDI):

Why was it needed:

A significant technological advancement for the GAAFET technology has been achieved through the creation of a fully isolated "bottom dielectric" (BDI) area under each transistor. The

need for this technology arose from the "fat fin" effect (also referred to as sub-fin parasitic channel leakage) that is unique to the GAAFET architecture. Due to the fact that the GAAFET nanosheet stack is placed atop a bulk silicon substrate (or upon a silicon pedestal above the substrate), the gate metal that fills the intersheet gaps extends downward below the bottom nanosheet to reach down to the substrate region. As a result, a parasitic transistor channel develops between the substrate and gate metal, leading to significant off-state leakage currents. Unlike FinFETs, this problem is present at a greater extent of severity for GAAFETs and necessitates the development of an entirely novel isolation module. Parasitic channels are an important issue in scaled CMOS technologies, and the suppression of parasitic channels in nanosheet GAAFET transistors is very difficult and significantly more complicated than for FinFETs. Solutions include using a heavily doped substrate punch-through stopper (PTS) and by adding an insulating dielectric layer for bottom isolation, which is termed the BDI module.

Process flow and nanofabrication techniques:

"The BDI module begins during epitaxial stack growth, where the bottom SiGe layer is grown with a higher germanium concentration than the upper sacrificial SiGe layers. This compositional difference creates an etch rate differential that allows the bottom layer to be selectively removed independently of the upper stack. After stack patterning and fin definition the high Ge bottom layer is etched out, and the resulting cavity is backfilled with a low k dielectric such as silicon oxynitride via ALD or CVD. This forms a continuous insulating floor between the substrate and eliminates the parasitic sub-fin conduction channel." [42]

Technological advancement and yield impact:

Full bottom dielectric isolation provides immunity from sub-channel leakage due to process variation from the parasitic fat fin structure unique to nanosheet GAAFET and enables power performance co-optimization. "A full BDI scheme demonstrates an 15.2% increase in speed and 18.9% less power consumption which is a vast improvement in performance compared with punch-through stopper approaches." [42] In terms of yield, BDI eliminates sensitivity to source/drain recess depth variability (a well-known process variability), since dielectric isolation is unaffected by small recess depths in contrast to PTS doping. Additionally, however, BDI adds additional process steps; it requires fabrication of a bottom SiGe layer having a high Ge fraction, but which exhibits independent etching selectivity; and finally, it necessitates modifications to the release chemistry used for the upper SiGe layers so that they remain compatible with the decreased Ge concentration of those layers.

Conclusion

The trajectory of the development of transistor manufacturing processes from planar CMOS through FinFET to finally the Gate-All-Around nanosheet architecture represents one of the longest-sustained, yet arguably most complex, engineering endeavors in the history of manufacturing. Each architectural generation was not based on preference for a particular theory; however, it was based on the necessities of the fabrication limitations of the prior process technologies. Therefore, the fabrication needs of the previous generation defined the requirements for developing the next generation of architectures.

The manufacturing discipline established by planar CMOS fabrication continues to govern the practices of the entire semiconductor industry today. STI provided a solution to the

isolation scaling problems presented by LOCOS. Additionally, since STI relied on CMP planarization to provide a planar surface, planarity became an essential component of all subsequent lithographic processing steps. Thermal gate oxidation and self-aligned polysilicon gate patterning together created a self-aligned transistor structure that allowed for the mass-production of large numbers of integrated circuits at reasonable yields and costs. Absent these fundamental techniques, there would be no present-day cost and yield economics for producing large quantities of functional transistors used in integrated circuits today.

FinFETs extended the semiconductor industry's ability to continue scaling for another decade beyond what can be achieved using planar geometries. By demonstrating that lithography resolution limits do not need to be insurmountable obstacles, but rather they are simply process-engineering challenges, SADP and SAQP allowed for the creation of features at pitches that were not directly resolvable due to the wavelength limitations of the exposure tool used for lithography. HKMG removed the "gate-leakage wall" that resulted from SiO2 thickness scaling. This allowed the gate dielectric to once again become a controlled process variable, rather than a fixed physical limitation. Using high-aspect ratio anisotropic RIE, the 3-D fin geometry allows for electrostatic control over three sides of the gate versus just one side in traditional planar structures. This directly improves sub-threshold operation and decreases leakage current at nodes that would otherwise be non-operational. Collectively, these advances enabled the 22 nm, 14 nm, 10 nm, and 7 nm process nodes that underlie many of the processors, graphics chips, and SoCs found in mobile systems and other areas that constitute much of today's computing infrastructure.

GAAFET nanosheet architecture has similarly continued this trajectory even longer. With Samsung's 3 nm GAA process going into production in 2022 and TSMC and Intel working toward their own nanosheet-based nodes at 2 nm and below, the fabrication innovations necessary to create GAAFET epitaxial superlattice growth, selective-release etching with greater than 100:1 selectivity, inner-spacer integration, conformal ALD gate deposition in sub-10 nm enclosed spaces, and bottom-dielectric isolation are examples of fabrication complexity that were previously thought to be outside the realm of manufacturable technology as little as ten years ago. Each of these techniques contributes directly to both the yield and performance characteristics of devices that are now being fabricated at commercial volumes. Improved electrostatic control afforded by the nanosheet gate results in reduced operating voltage levels, decreased leakage-power consumption, and increased transistor densities. All of these result in quantifiable improvements in energy efficiency and computational throughput in chips utilized in global data centers, AI-accelerator hardware, and consumer-electronics applications.

Similarly significant in terms of their actual-world impacts are the yield improvements realized with each successive generation. Pattern-loading solutions developed for STI-CMP influenced the dummy-fill layouts and design rules governing all advanced chip designs today. CDU/LER control techniques developed for SADP/SAQP fin-patterning account for the threshold-voltage uniformities allowing billions of transistors per chip to be manufactured at commercially acceptable yields. The multi-step channel-release technique developed to eliminate mechanical distortion resulting from GAAFET fabrication is a clear illustration of how atomic-level process engineering translates into practical, ship-able silicon. Every yield-related challenge solved during research-and-development stages lowers the cost-per-functionally-useful transistor at commercial production stages. Lowering this cost-per-transistor ultimately

determines if a specific technological node is economical enough to move out of a laboratory environment and into commercial production.

In addition to establishing the technical foundation for future post-GAAFET device concepts currently being explored, the fabrication innovations developed across these three architectural generations have also established a wealth of process knowledge that precedes them. For example, complementary FET (CFET) architecture depends on stacking two complementary MOSFETS (PMOS/NMOS) vertically in the same foot-print area. CFET architecture is dependent on several key fabrication techniques developed specifically for GAAFET including epitaxial stack formation, selective etching, and conformal ALD deposition. Similarly, two-dimensional channel-materials such as Molybdenum Disulphide (MoS2) and Transition Metal Dichalcogenides (TMDs), which are presently under investigation as alternatives to Silicon channels at sub-1 nm EOT (effective oxide thickness), depend on identical atomic-level control over ALD and selective etching that existed during nanosheet fabrication. As such, while historic, this body of process knowledge is also entirely relevant as the direct basis of precedent upon which future generations of transistor architectures will be built.

This documented report illustrates that semiconductor scaling is fundamentally a fabrication issue. Device architecture follows wherever process innovation provides sufficient capabilities to allow. Therefore, continued advancements in computing capability whether in AI, HPC (high-performance computing), mobile technology or elsewhere remain reliant on continuous development of those nanofabrication methods that provide sufficient capabilities to manufacture each succeeding generation of transistors at acceptable yields.

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